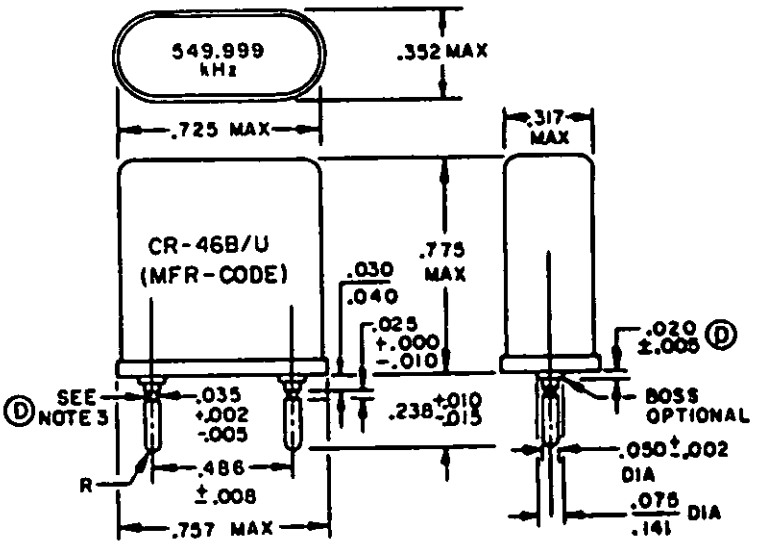
# MILITARY SPECIFICATION SHEET

## CRYSTAL UNIT, QUARTZ, CR-46B/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 200 to 550 kHz; fundamental; noncontrolled; antiresonance.




| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .002 | .05 | .020 | .51 | .050 | 1.27 | .352 | 8.94 |
| .005 | .13 | .025 | .64 | .075 | 1.91 | .486 | 12.34 |
| .008 | .20 | .030 | .76 | .141 | 3.58 | .725 | 18.42 |
| .010 | .25 | .035 | .89 | .238 | 6.05 | .757 | 19.23 |
| .015 | .38 | .040 | 1.02 | .317 | 8.05 | .775 | 19.69 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
Ⓓ 3. The pin undercut may be omitted.

Ⓓ denotes change

FSC 5955



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TABLE I. Requirements.

| Requirement | Specified value or condition |
|---|---|
| Holder | HC-6/U (MIL-H-10056/2) |
| Dimensions, marking, and configuration | See figure |
| Frequency range | 200 to 550 kHz, inclusive |
| Frequency tolerance (operating temperature range) | ±100 PPM |
| Equivalent resistance | See table III |
| Mode of oscillation | Fundamental |
| Antiresonance, load capacitance | 20 ±0.5 pF |
| Operating temperature range | -40° to +85°C, inclusive |
| Test set, calibration values, and rated drive level | See table II |
| Shock (specified pulse): | |
| Frequency change permitted | ±10 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Vibration: 1/ | |
| Frequency change permitted | ±10 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Temperature run: | |
| Frequency change permitted | ±10 PPM |
| Equivalent resistance change permitted | ±15 percent |
| Bond strength | kHz — Grams, minimum<br>200 to 250 — 700<br>250+ to 320 — 500<br>320+ to 370 — 400<br>370+ to 435 — 300<br>435+ to 550 — 250 |

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (total excursion 0.030 inch).

TABLE II. Test set, calibration values, and rated drive level.

| Government reference standard test set | Band of specified frequencies, inclusive | Calibration values | | | Rated drive level |
|---|---|---|---|---|---|
| | | Resistance | Crystal current | Resistor voltage drop | |
| | kHz | Ohms | mA | Volt | mW |
| TS-710/TSM | 200 to 225 | 3700 | --- | 2.72 | 2.0 ±0.4 |
| | 225+ to 275 | 4200 | --- | 2.90 | |
| | 275+ to 325 | 4600 | --- | 3.03 | |
| | 325+ to 375 | 4900 | --- | 3.13 | |
| | 375+ to 425 | 5300 | --- | 3.26 | |
| | 425+ to 475 | 5600 | --- | 3.35 | |
| | 475+ to 500 | 6000 | --- | 3.47 | |
| | 500+ to 550 | 3500 | --- | 2.65 | |



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TABLE III. Equivalent resistance.

| Frequency range, inclusive | Maximum resistance |
|---|---|
| kHz | Ohms |
| 200 to 225 | 5,300 |
| 225+ to 275 | 6,000 |
| 275+ to 325 | 6,500 |
| 325+ to 375 | 7,000 |
| 375+ to 425 | 7,500 |
| 425+ to 475 | 8,000 |
| 475+ to 500 | 8,500 |
| 500+ to 555 | 5,000 |

NOTES:

(D) 1. The socket designed to mate with crystal holder HC-6/U is Part No. M12883/04-01 (see MIL-S-12883/4).
For Navy use, other sockets listed in MIL-STD-242 may be used.

2. This crystal-unit type is to be used for special applications and for replacement purposes only. use. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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